Practice Quiz: Mensuration, Ages, and Simplification

Mathematics Workshop

Part 1: Mensuration

1. The length of a rectangle is 20% more than its breadth. If the area of the rectangle is 480 sq cm, find its perimeter.

A) 88 cm
B) 92 cm
C) 96 cm
D) 104 cm

2. The area of a circle is 154 sq cm. Find the length of the side of a square whose perimeter is equal to the circumference of the circle.

A) 11 cm
B) 22 cm
C) 44 cm
D) 7 cm

3. A wire is in the form of a circle of radius 42 cm. If it is bent into a square, what will be the area of the square? (Use $\pi = \frac{22}{7}$)

A) 4356 sq cm
B) 4489 sq cm
C) 3600 sq cm
D) 5184 sq cm

4. The diagonal of a cube is $6\sqrt{3}$ cm. Find its total surface area.

A) 144 sq cm
B) 216 sq cm
C) 108 sq cm
D) 432 sq cm

5. If the radius of a sphere is doubled, what is the ratio of the volume of the original sphere to that of the new sphere?

A) $1 : 2$
B) $1 : 4$
C) $1 : 8$
D) $1 : 16$

6. The base of a triangle is 15 cm and height is 12 cm. The height of another triangle of double the area having the base 20 cm is:

   A) 9 cm
   B) 18 cm
   C) 12.5 cm
   D) 10 cm

7. A cylinder has a radius of 7 cm and a height of 10 cm. Find its curved surface area.

   A) 220 sq cm
   B) 440 sq cm
   C) 660 sq cm
   D) 1540 sq cm

8. The perimeter of a rhombus is 40 cm and one of its diagonals is 12 cm. Find the area of the rhombus.

   A) 48 sq cm
   B) 96 sq cm
   C) 72 sq cm
   D) 64 sq cm

9. If the length, breadth, and height of a cuboid are increased by 10% each, the percentage increase in volume is:

   A) 30%
   B) 33.1%
   C) 21%
   D) 11%

10. The radius of a cone is 3 cm and its slant height is 5 cm. Find its volume.

   A) 12π cu cm
   B) 15π cu cm
   C) 20π cu cm
   D) 36π cu cm

Part 2: Problems on Ages

11. The ratio of the ages of Father and Son at present is $4 : 1$. After 20 years, the ratio becomes $2 : 1$. What is the present age of the Father?

   A) 30 years
   B) 40 years
   C) 50 years
   D) 60 years

12. Six years ago, the ratio of the ages of Kunal and Sagar was $6 : 5$. Four years hence, the ratio of their ages will be $11 : 10$. What is Sagar's age at present?

A) 16 years
B) 18 years
C) 20 years
D) 22 years

13. A man is 24 years older than his son. In two years, his age will be twice the age of his son. The present age of the son is:

A) 14 years
B) 18 years
C) 20 years
D) 22 years

14. The sum of the present ages of a mother and her daughter is 60 years. Twelve years ago, the mother was eight times as old as her daughter. How old is the daughter now?

A) 16 years
B) 12 years
C) 18 years
D) 20 years

15. Rajan's age is 3 times that of his son. Five years ago, his age was 4 times that of his son. What is Rajan's present age?

A) 45 years
B) 50 years
C) 35 years
D) 60 years

16. The ages of A and B are in the ratio $5 : 7$. If the difference between their ages is 8 years, what is the sum of their ages?

A) 40 years
B) 48 years
C) 56 years
D) 64 years

17. At present, the ratio of the ages of Maya and Chhaya is $6 : 5$ and fifteen years from now, the ratio will get changed to $9 : 8$. Maya's present age is:

A) 21 years
B) 24 years
C) 30 years
D) 40 years

18. The average age of a group of 10 students is 15 years. When a teacher's age is included, the average increases by 1. What is the teacher's age?

A) 25 years
B) 26 years

C) 31 years

D) 35 years

19. A person was asked to state his age in years. His reply was, "Take my age three years hence, multiply it by 3 and then subtract three times my age three years ago and you will know how old I am." What is the age of the person?

A) 18 years

B) 20 years

C) 24 years

D) 32 years

20. Ten years ago, P was half of Q in age. If the ratio of their present ages is $3 : 4$, what is the sum of their present ages?

A) 35 years

B) 30 years

C) 40 years

D) 45 years

Part 3: Simplification

21. Evaluate: $5005 - 5000 \div 10$

A) 0.5

B) 4505

C) 50

D) 4995

22. Simplify: $b - [b - (a + b) - \{b - (b - \overline{a - b})\} + 2a]$

A) a

B) $2a$

C) $4a$

D) 0

23. Find the value of: $\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}$

A) 4

B) 6

C) 8

D) 10

24. Calculate: $(256)^{0.16} \times (256)^{0.09}$

A) 4

B) 16

C) 64

D) 256.25

25. Solve: $\frac{(0.05)^2+(0.41)^2+(0.073)^2}{(0.005)^2+(0.041)^2+(0.0073)^2}$

A) 10

B) 100

C) 1000

D) 0.1

26. If $x+\frac{1}{x}=5$, then the value of $x^2+\frac{1}{x^2}$ is:

A) 23

B) 25

C) 27

D) 30

27. Simplify: $10.8 \times 5.5 - 12.5$

A) 46.9

B) 48.2

C) 50.5

D) 45.4

28. What is the value of $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}$?

A) $15/16$

B) $13/16$

C) 1

D) $7/8$

29. Find x if $2^{x-1}+2^{x+1}=320$.

A) 5

B) 6

C) 7

D) 8

30. Simplify: $(a-b)^3+(b-c)^3+(c-a)^3$ where $a+b+c=0$ is not required, given as a general identity.

A) $3(a-b)(b-c)(c-a)$

B) $a^3+b^3+c^3$

C) 0

D) $3abc$